

Neural Control & Coordination

SHORT NOTES

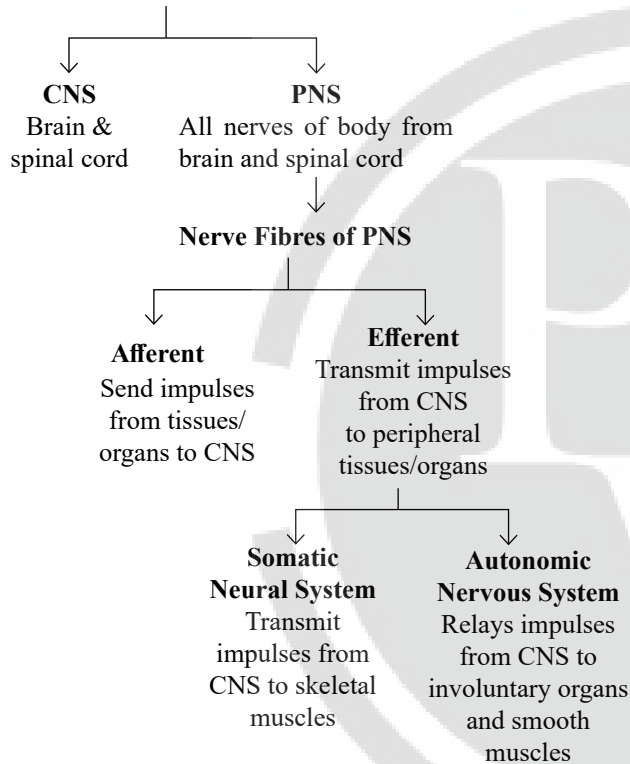
Introduction

- ❖ The neural system and the endocrine system jointly coordinate and integrate all the activities of the organs so that they function in a synchronised manner.

Neural System

- ❖ Neurons can detect, receive & transmit stimulus.
- ❖ *Hydra* – Network of neurons.
- ❖ Insects – Organised neural system with brain and ganglia.
- ❖ Vertebrates – Well developed neural system.

HUMAN NEURAL SYSTEM



Visceral Nervous System (VNS):

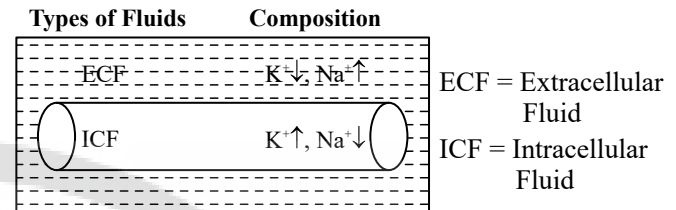
- Part of Peripheral Nervous System (PNS).
- Complex of nerves, fibres, ganglia and plexuses.

Neuron

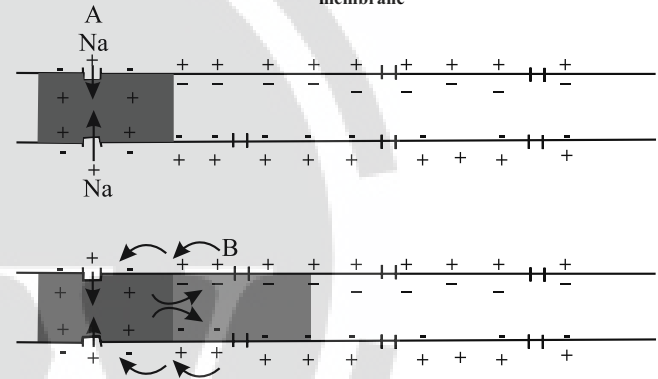
- ❖ Neuron is the structural and functional unit of the neural system.
- ❖ Composed of a cell body, dendrites and axon.
 - Nissl's granules are present.
- ❖ Three types of neurons: Multipolar, bipolar and unipolar.
- ❖ Cell body contains cell organelles except centriole.
- ❖ Axons can be myelinated or non-myelinated.
 - + CNS → Oligodendrocytes → Myelination
 - + PNS → Schwann cells → Myelination

Generation and Conduction of Nerve Impulse

- Axon Membranes (selectively permeable)**
- More permeable for K^+
 - Nearly impermeable to Na^+
 - Impermeable to negatively charged proteins
 - Sodium-potassium pump maintain ionic gradients



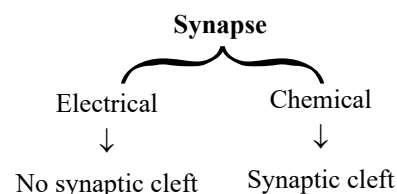
- ❖ Flow of charge is from A $\xrightarrow[\text{Outside membrane}]{\text{Inside membrane}}$ B.



Transmission of Impulse

- ❖ Done with the help of synapse.
- ❖ **Events:**
 - + Neurotransmitters released in synaptic cleft.
 - ↓
 - + Bind to receptors on post synaptic neuronal membrane.
 - ↓
 - + Opening of ion channels in post synaptic membrane.
 - ↓
 - + Generates a new potential in post synaptic membrane.

Synapse



Central Nervous System (CNS)

Protective Coverings

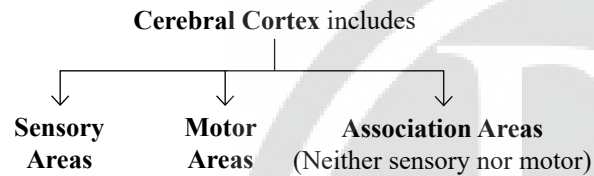
- Skull
- Meninges

	Outer	Middle	Inner
Name of Meninx	Dura mater	Arachnoid (Thin)	Pia mater
In Contact with	Skull		Brain

Brain

Forebrain

- ❖ Cerebrum → Cerebral hemispheres are connected by corpus callosum.
- ❖ Thalamus.
- ❖ Hypothalamus.
- ❖ Limbic system → Complex structure involve amygdala, hippocampus, inner part of cerebral hemisphere and hypothalamus.



Midbrain

- ❖ Corpora Quadrigemina → 4 lobes on dorsal side.
- ❖ Cerebral Aqueduct → Canal passes through midbrain.
- ❖ Brain Stem: Midbrain, pons and medulla oblongata.

Hindbrain

- ❖ Pons
- ❖ Cerebellum
- ❖ Medulla oblongata

Spinal Cord

- ❖ Through foramen of magnum aperture the spinal cord passes out of skull.
- ❖ Neurocoel - Cavity of spinal cord.
- ❖ 2 types of matter

Grey Matter

- ❖ Present inside
- ❖ H-shaped
- ❖ Butterfly shaped

White Matter

- ❖ Surrounds the grey matter